Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 8

1. a) (2 points): Using Pascal's identity, prove that:

$$\binom{n+1}{2} = \binom{n}{2} + n$$

b) (8 points): Using part a) repeatedly, prove that

$$1 + 2 + 3 + \ldots + (n-1) + n = \frac{n(n+1)}{2}$$

For the avoidance of doubt, using a rainbow method proof will get you 0 points.

Solution:

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2. How many pairs of positive integers x, y satisfy $x^2 = 60 + y^2$? Prove your answer.

Solution:

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3. You have a bunch of congruent (identical) equilateral triangles. Each triangle has three numbers on it: 1, 2, and 3 at different vertices. Is it possible to make a pile of these triangles exactly on top of each other so that the sum of numbers at each vertex is 55? Prove your answer.

Solution:

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4. Is it possible to tile a 10x10 square with 25 tetrominoes that look like this?  (For the avoidance of doubt, you are allowed to rotate the pieces in four different ways). Prove your answer.

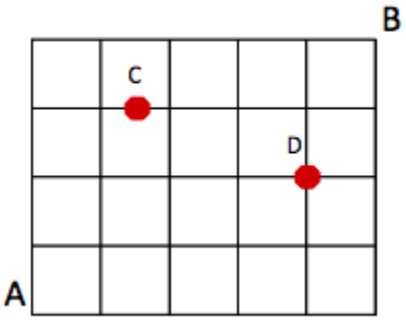
Solution:

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5. A city consists of a 4x5 grid of blocks. Roman starts at the bottom left corner and is trying to get to the top right corner, by going only up and to the right. (For the avoidance of doubt, he will be walking exactly 9 blocks).

a) (2 points) How many ways are there for Roman to do so?

b) (2 points) The city now closes the intersection marked D. How many of Roman's paths from A to B in part a) went through D?



c) (3 points) How many ways are there for Roman to get from A to B without going through point D?

d) (3 points) Now, in addition to closing the intersection marked D, the city also closes the road marked C. How many ways are there for Roman to get from A to B without going through either the closed intersection or the closed road?

Solution:

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6. In how many ways can one read the word COMBINATORICS starting from the upper left corner and going either right or down? Prove your answer.

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C O M B I N A T O R I C S
O M B I N A T O R I C S
M B I N A T O R I C S
B I N A T O R I C S
I N A T O R I C S
N A T O R I C S
A T O R I C S
T O R I C S
O R I C S
R I C S
I C S
C S
S
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Solution: